

■ UNSW × MISTRAL AI × ATlassian — 2026

SPATIAL BRAIN

A research paper becomes a Jira issue becomes a pull request — and every hop stays visible on one canvas.

■ THE OFFICIAL PROBLEM STATEMENT

“How might AI help multi-disciplinary teams make sense of information, present ideas, align on decisions, and review work more effectively?”

■ WHAT THE PROBLEM INCLUDES

DESIGN ENGINEERING PRODUCT OPERATIONS BUSINESS LAW SCIENCE
PULL REQUESTS WHITEBOARD PHOTOS RECORDINGS DOCUMENTS PDFS SPREADSHEETS RAW DATA

 WHO THIS IS FOR

PRODUCT MANAGER

Priya Raman

RESEARCH SCIENTIST

Dr Aisha Khan

SOFTWARE ENGINEER

Marco Silva

The pain: Cross-disciplinary work loses its reasoning in the handoff. A scientist's finding is retyped into a ticket, the ticket loses the constraint that made it necessary, and the engineer implements something subtly wrong.

AI should package and pressure-test context — not replace judgment.

■ THE SOLUTION, ONE SENTENCE

A shared semantic canvas where multi-modal ingest and a Mistral specialist roster cover make-sense / present / align / review — while Spatial Brain stays the system of record.

```
Canvas (Next.js) — FastAPI — SQLite (system of record)
|
├─ Multimodal extract → candidates → human promote
├─ Align / Present / Review graph tools
├─ Mistral Agents (Coordinator + specialists)
├─ Lineage assembler → MCP tools → coding agent
└─ Jira outbound + GitHub/Jira inbound webhooks
```

THE SOLUTION, ONE SENTENCE

FOUR PILLARS — MAPPED TO THE PROBLEM STATEMENT

MAKE SENSE

PDF / whiteboard photo / CSV → candidates → human promote

PRESENT IDEAS

Present Mode + stakeholder brief + generated one-pager

ALIGN ON DECISIONS

Check alignment + decision trail — decided / deferred / rejected

REVIEW WORK

Jira assign + PR badge + constraint checklist + inbound webhooks

+ RECOMMEND

From a finding or constraint, Mistral auto-creates linked task nodes

- Human review / confirmation / reassignment
- Human review / confirmation / reassignment
- Human review / confirmation / reassignment

■ LIVE DEMO, SIX BEATS

- 1 Science · PDF — make sense**
Drop the paper. Mistral proposes findings and constraints; promote a few.
- 2 Design · whiteboard — make sense**
Drop a photo of a sketch. Same review UI, different input.
- 3 Ops · spreadsheet — make sense**
Drop a CSV. The data analyst path proposes metric constraints.
- 4 Product · align — align on decisions**
Check alignment on the recovery task — the Arbiter flags the OTP-vs-magic-links contradiction.
- 5 All · present — present ideas**
Present Mode walks claim → quote → implication → task, with a generated one-pager.
- 6 Engineering · review — review work**
Assign to Marco → real Jira issue. Link a PR. Run the constraint checklist.

WHY WE'RE DIFFERENT

HEPTABASE

Human-curated notes with AI assistance layered on top. No agent-facing API — the knowledge stays inside the app.

AGENT-MCP

Built for AI coding agents collaborating in one codebase. Developer-only, not multi-disciplinary humans.

ATLASSIAN'S JIRA MCP

Exposes *existing* Jira data to agents. Doesn't generate the reasoning behind a ticket in the first place.

Spatial Brain: AI extracts automatically, with citations. Multi-disciplinary humans *and* agents. Generates the context, then writes it into Jira — upstream of every competitor listed here.

■ FOUR CLAIMS WE CAN DEFEND LIVE

01 · LINEAGE

A graph walk, not an LLM guessing. Plain BFS in `lineage.py` — Mistral only summarizes a set that's already deterministically selected.

02 · PROVENANCE

Every extracted fact carries a verbatim quote, page number, confidence score, and whether a human or a model asserted it.

03 · THE LOOP CLOSSES

An agent doesn't just read context via MCP — it writes back. A reported PR shows up as a badge on the canvas and a Jira comment.

04 · REAL INTEGRATIONS

A real Jira Cloud issue via actual REST calls, with ambiguous-timeout handling — not a mock, not a screenshot.

■ SCORED AGAINST THE OFFICIAL RUBRIC

30% Value + Human Insight	30% Technical Execution	20% Creativity + Design	20% Feasibility + Scalability
HIDDEN SIGNALS AWARD — “Uncover a non-obvious pattern, contradiction, anomaly, or connection.” That’s <i>exactly</i> what Check Alignment does, live, on stage.			

■ CLOSING

**Nothing here is a new place to work.
It is the missing edge between the places people already work.**

Spatial Brain — Priya, Dr Aisha Khan, Marco & team.

UNSW x Mistral AI x Atlassian, 2026.